

Pradyuman

Backend & Systems Engineer — Go, Rust, C++ — Distributed Systems, Storage, Networking

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SUMMARY

Senior Backend & Systems Engineer (4+ YOE) designing and scaling high-throughput, fault-tolerant services using Go and Rust. Deep experience in large-scale systems design, observability (OTel), and storage technologies (ClickHouse, Object Store), with a focus on performance and reliability ownership.

SKILLS

Languages: Go, Rust, C++, Python, Bash

Systems & Internals: Concurrency & Thread Scheduling, Syscalls, File Systems (ext4, NFS), Socket Programming

Distributed & Storage Systems: ClickHouse, Redpanda, Cassandra, Object Store, Replication, Partitioning

Observability Stack: OpenTelemetry (Traces/Logs/Metrics), Fluent Bit, Prometheus, Grafana, Telemetry Pipelines

Infrastructure & Cloud: Linux, Docker, Kubernetes, Terraform, Helm, GitHub Actions, AWS, GCP, Azure

Core Engineering Skills: System Design, API Design, High-Availability Systems, Performance Optimization, DSA

EXPERIENCE

CtrlB

April 2024 – Present

Senior Software Engineer → Lead Software Engineer

Bengaluru, Karnataka

- Contributing to the core data engine written in Rust — implementing new features, optimizing query execution, and improving query latency by 30% while reducing crash frequency under peak load.
- Designed and engineered backend services in Go for authentication, RBAC, and multi-tenant data isolation, enabling a secure, unified lake architecture across multiple clients.
- Built and maintained the observability control plane to manage OpenTelemetry agents, automating configuration and deployment across over 100 nodes, eliminating about 80% of manual operations.
- Deployed a self-hosted Kubernetes cluster from scratch, reducing developer onboarding time from 4–5 hours to under 15 minutes through automated provisioning.
- Led and mentored a 3-member engineering team, conducting design and code reviews, driving architectural decisions, and increasing feature delivery rate by 25% through improved review and CI processes.

Middleware

Sept 2023 – Feb 2024

Senior Software Engineer

Ahmedabad, Gujarat

- Designed modular deployment workflows using Helm and Kubernetes manifests, enabling rapid agent provisioning across multi-tenant clusters.
- Integrated observability pipelines with ClickHouse, Redpanda, and Cassandra to achieve low-latency ingestion and high query throughput.
- Automated CI/CD workflows via GitHub Actions, improving release reliability and reducing manual deployment time by 40%.

Gap Inc.

June 2021 – Sept 2023

Software Engineer

Hyderabad, Telangana

- Drove OpenTelemetry adoption across a large-scale AKS deployment, building a vendor-neutral observability layer for 100+ microservices.
- Migrated infrastructure from RHEL to OEL, saving \$50K annually while maintaining SLA-grade availability and performance.
- Authored Chef cookbooks to standardize and harden VM provisioning, enhancing baseline security and configuration consistency.

PROJECTS

Mini-NFS Protocol Implementation | Rust | 🐙

Systems Project

- Built a custom NFS-like protocol over TCP to support LIST, READ, and WRITE operations with emphasis on concurrency and protocol correctness.
- Built the file server and client components using Rust, implementing line-based command parsing and streaming byte-level I/O for reliable remote file access.

Mini Object Store (S3-like) | Golang | 🐙

Storage Systems Project

- Developing a lightweight, S3-compatible object store supporting core operations—PUT, GET, and LIST.
- Implements a simple on-disk layout and metadata index for object tracking and efficient lookups.

EDUCATION

National Institute of Technology(NIT), Hamirpur

Bachelor of Technology in Computer Science

2017 - 2021

Hamirpur, HP